

Phase 5 — Final

Introduction

This phase focuses on evaluating and improving the proposed methodology by comparing it with recent research work. The objective is to identify gaps in existing approaches and demonstrate how the current project provides a better solution for predicting song popularity.

Review of Recent Research

Research Paper 1

Title: Popularity Prediction of Music by Machine Learning Models

- Machine learning models such as XGBoost and Linear Regression were used.
- The study achieved moderate accuracy but did not include advanced feature engineering techniques.
- Link:
https://www.researchgate.net/publication/370712842_Popularity_Prediction_of_Music_by_Machine_Learning_Models

Research Paper 2

Title: Predicting Song Popularity in the Digital Age

- Focused on regression-based analysis using Spotify features.
- Identified key factors such as danceability and duration.
- Limited in terms of model diversity and feature enhancement.
- Link:
https://www.researchgate.net/publication/381621689_Predicting_song_popularity_in_the_digital_age_through_Spotify_data

Research Paper 3

Title: Predicting Song Popularity Using Spotify Audio Features (2023)

- Used Random Forest and Extra Trees models.

- Achieved good accuracy but relied only on existing dataset features.
- Link: <https://www.tandfonline.com/doi/full/10.1080/23270012.2023.2239824>

Research Paper 4

Title: Exploring Spotify Music Popularity with Machine Learning (2024)

- Applied machine learning to analyze patterns in music data.
- Highlighted the importance of feature relationships but lacked clustering techniques.
- Link: https://www.researchgate.net/publication/385516062_Exploring_Spotify%27s_Music_Popularity_Dynamics_and_Forecasting_with_Machine_Learning

Research Paper 5

Title: Music Recommendation System Using Deep Learning (2024)

- Used deep learning models for prediction.
- Achieved high accuracy but required high computational resources.
- Link: https://www.researchgate.net/publication/386149700_Music_Recommendation_System_on_Spotify_Using_Deep_Learning

Comparison with Proposed Methodology

Most existing research focuses on basic machine learning models or deep learning approaches without enhancing the dataset itself.

In contrast, this project introduces several improvements:

- Use of multiple models (Linear Regression, Decision Tree, LightGBM)
- Application of **feature engineering** to create new meaningful features
- Use of **KMeans clustering** to identify hidden patterns
- Proper **data preprocessing and standardization**

Novelty and Contribution

This project fills an important gap by improving the dataset representation rather than relying only on complex models.

Unlike existing research, this work combines feature engineering, clustering, and multiple machine learning models to enhance prediction performance. This approach allows better understanding of relationships between features and improves the accuracy of predicting song popularity.

Results and Findings

The implementation of multiple models showed that:

- Linear Regression provided a baseline performance
- Decision Tree captured non-linear relationships
- **LightGBM achieved the best performance** in terms of R2 score and RMSE

This confirms that the proposed methodology is effective.

Conclusion

The comparison with recent research demonstrates that this project provides a more comprehensive approach by combining data preprocessing, feature engineering, clustering, and multiple machine learning models. This results in improved prediction accuracy and better understanding of music data patterns.

◆ References

[1]
https://www.researchgate.net/publication/370712842_Popularity_Prediction_of_Music_by_Machine_Learning_Models

[2]
https://www.researchgate.net/publication/381621689_Predicting_song_popularity_in_the_digital_age_through_Spotifys_data

[3] <https://www.tandfonline.com/doi/full/10.1080/23270012.2023.2239824>

[4]

https://www.researchgate.net/publication/385516062_Exploring_Spotify%27s_Music_Popularity_Dynamics_and_Forecasting_with_Machine_Learning

[5]

https://www.researchgate.net/publication/386149700_Music_Recommendation_System_on_Spotify_Using_Deep_Learning